

- (d) The variation of the force F acting on the spring with the compression x of the spring is shown in Fig. 3.2.

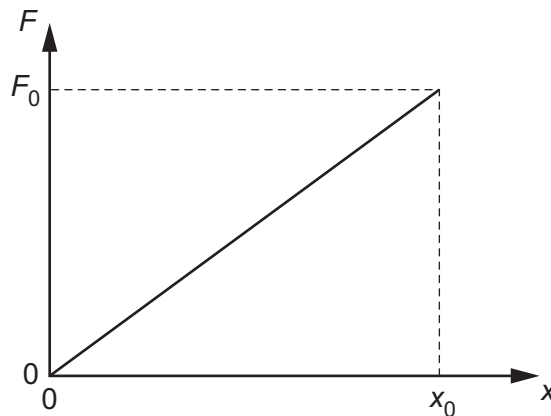


Fig. 3.2

Use the information in (b) and your answer in (c) to show that the maximum force F_0 exerted on the spring by the block is 700 N.

[2]

- (e) Use the information in (d) to determine, for the instant that the block is first brought to rest by the spring, the magnitude of:

- (i) the resultant force acting on the block

resultant force = N [2]

- (ii) the acceleration of the block.

acceleration = ms^{-2} [2]

[Total: 11]

